

ESP32-Based Environmental Data Logger with IoT Control and Cloud Backup

REPORT

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1. Abstract

This project presents an ESP32-based environmental data logger capable of real-time monitoring, automated control, and IoT-enabled remote access. It supports measurement of temperature, humidity, soil moisture, pressure, and light intensity and stores this data locally and on the cloud. The system integrates with Google Sheets and the Blynk platform for visualisation and remote control. Dual connectivity modes, utilising both WiFi and GSM, ensure continuous operation. Designed with scalability and reliability in mind, this device is suited for agriculture, research, and smart environment applications.

2. Introduction

The report provides the design and development of a real-time monitoring and control ESP32-based environmental data logger for environmental parameters. The system is suitable for use in agriculture, horticulture, and environmental studies and has automated control, robust local and cloud data logging, and complete remote access through IoT integration.

Objectives

- Measure temperature, humidity, soil moisture, atmospheric pressure, and light intensity.
- Record environmental data locally on a microSD card.
- Transfer sensor data to Google Sheets over the internet.
- Control connected devices automatically (e.g., pump, grow light) using sensor data.
- Allow remote monitoring and control through the Blynk IoT platform.
- Offer network redundancy by supporting WiFi and GSM-based data transmission.

System Overview

The data logger utilises the ESP32 microcontroller to acquire sensor readings at specified intervals. Data is shown locally, saved to an SD card, and uploaded to the cloud via WiFi or GSM (SIM800L). A real-time clock module stamps each reading. Both manual and automatic control of actuators through relay modules are supported by the system. Threshold values and operating modes can be set through the Blynk mobile app. The hardware assembly is enclosed in a custom 3D-printed case, designed for durability, sensor accessibility, and protection in semi-outdoor environments.

3. Project and System Details

3.1 Sensor Integration

- Temperature and Humidity Monitoring (DHT22)
- Soil Moisture Detection (Analog Sensor)
- Atmospheric Pressure Measurement (BMP180)
- Light Intensity Measurement (BH1750)

3.2 Data Logging

- Data is stored in CSV format on a microSD card.
- Cloud logging is performed via Google Sheets using HTTPS.
- Logging intervals are user-adjustable via the Blynk app.

Cloud logging in google sheets

3.3 Automated Control

- Automatic pump operation when soil moisture falls below a user-configured threshold.
- Grow light operations based on light intensity.
- Thresholds and automation modes are remotely configurable.
- Both relays have manual override.

3.4 Connectivity

- WiFi is the main network interface.
- GSM (SIM800L) as a failover if WiFi is not available.

- WiFiManager for dynamic WiFi credential configuration without code change.

3.5 User Interface

- The OLED display cycles through system status and sensor readings.
- Two buttons offer UI navigation and system reset.
- LED indicators indicate operation and connectivity status.

3.6 Software Architecture

The software is written based on the Arduino platform. It has a modular design for maintainability and readability. The most important tasks are allocated logically and make use of FreeRTOS where necessary for simultaneous/parallel execution.

3.6.1 Functional Modules

- Sensor reading and verification
- OLED display refresh and navigation
- SD card logging
- WiFi and GSM communication
- Blynk IoT interface
- Relay control logic
- Buck converter
- REAL Time Clock for time stamping (DS3231)

3.6.2 Cloud Integration

- Google Sheets receives data through HTTP POST requests via a Google Apps Script endpoint.
- The format consists of timestamp, temperature, humidity, soil moisture, light intensity, and pressure.
- Data is transmitted at configurable intervals with fallback logic to retry on failure.

3.7 Blynk IoT Platform

Real-time access to all sensor data and control commands for relays is provided through the Blynk app. The features include:

- Displaying live readings
- Specifying moisture and lux thresholds
- Enabling/disabling automation
- Toggling relays manually
- Adjusting logging intervals
- Forcing manual data uploads

3.8 Backup and Fail-Safes

For added system reliability, there are a number of fail-safes implemented:

3.8.1 SD Card Logging Backup

- When primary logging to the main CSV file is impossible, data is diverted to a temporary file on the same SD card.
- Logging resumes to the main file as soon as the problem is corrected.

3.8.2 Network Fallback

- The system tries to send data over WiFi initially.
- In case of failure to send via WiFi after a number of attempts, GSM is automatically set up and utilised to send data.

3.8.3 Logging Continuity

- Even if both cloud methods are not available, local logging proceeds uninterrupted.
- This prevents data loss even in poor network conditions.

3.8.4 System Recovery

- System or WiFi credentials can be reset with the help of buttons to reconfigure.
- All the errors and fallbacks are non-blocking and enable uninterrupted operation.

3.9 Operational Workflow

Startup:

- Initialises sensors, SD card, display, and connectivity modules.
- RTC time is verified and synchronised as required.

Loop Operation:

- Read all the sensors at the specified interval.
- Updates OLED display based on button interaction.
- Applies automation logic to control relays.
- Logs data to SD card.
- Sends data to Google Sheets via WiFi or GSM.
- Updates values to Blynk.

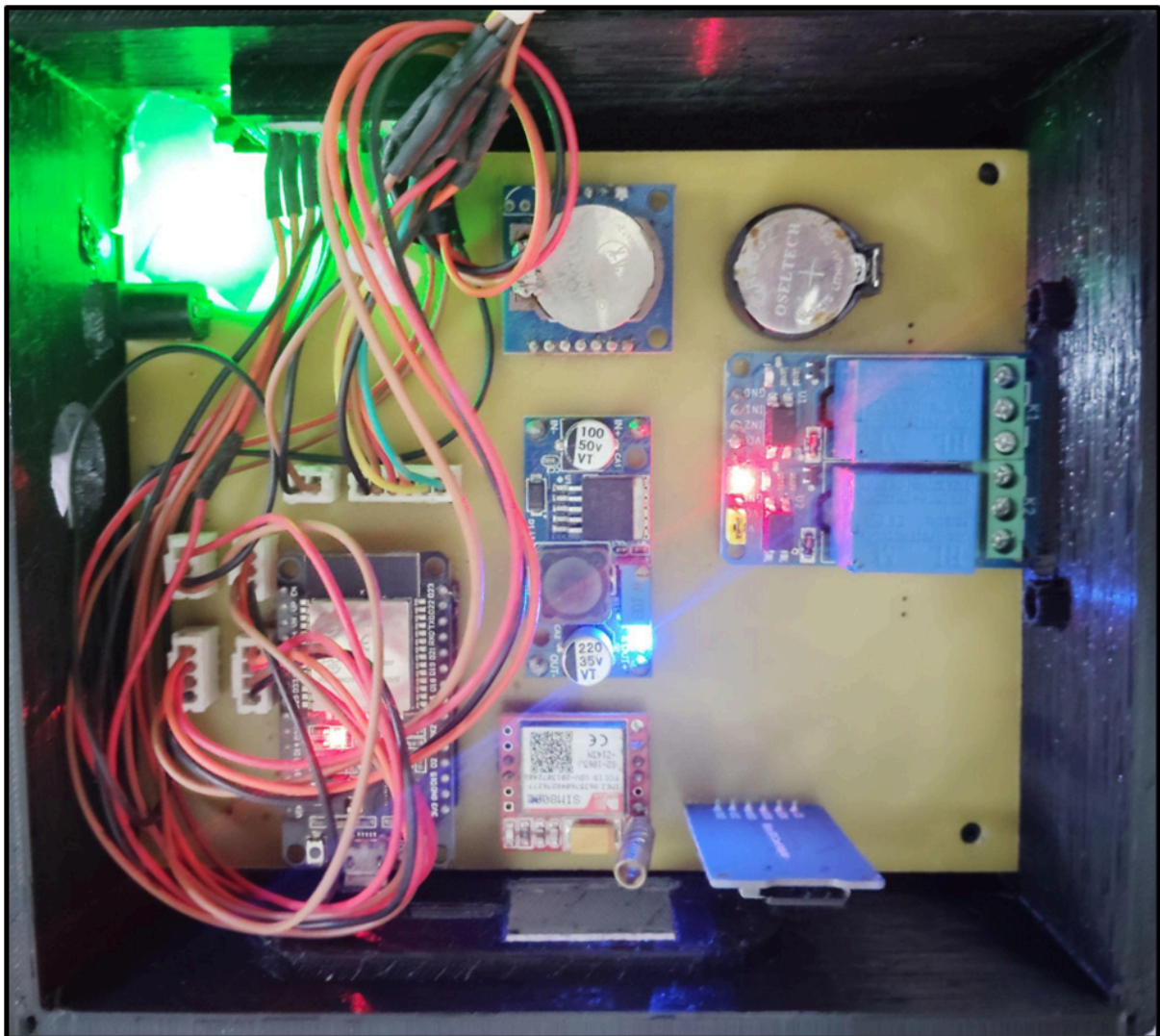
Manual Functions:

- Relay control and threshold changes via Blynk.
- Force logging and reset via button combinations.

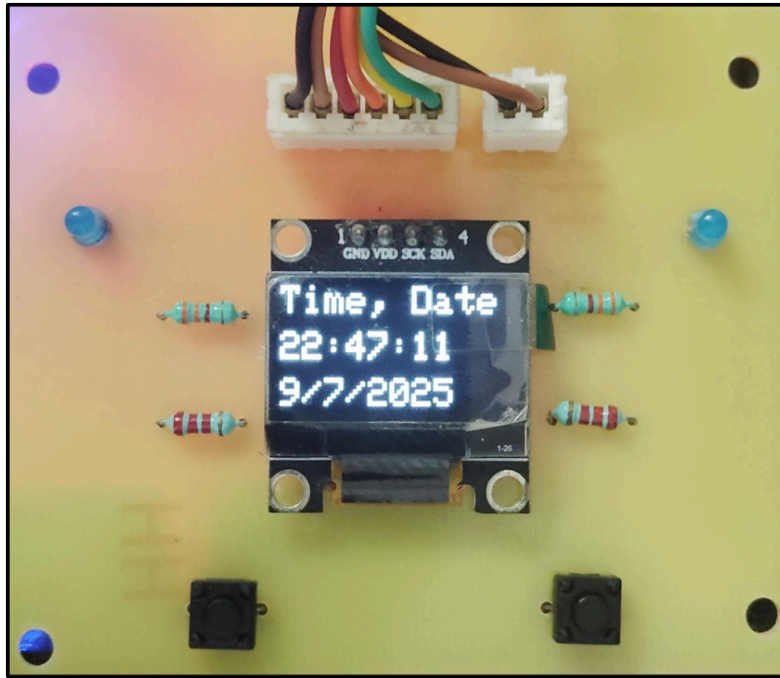
3.10 Applications

- Precision Agriculture
- Greenhouse Automation
- Weather Station
- Remote Environmental Monitoring
- Educational and Research Projects

3.11 Images



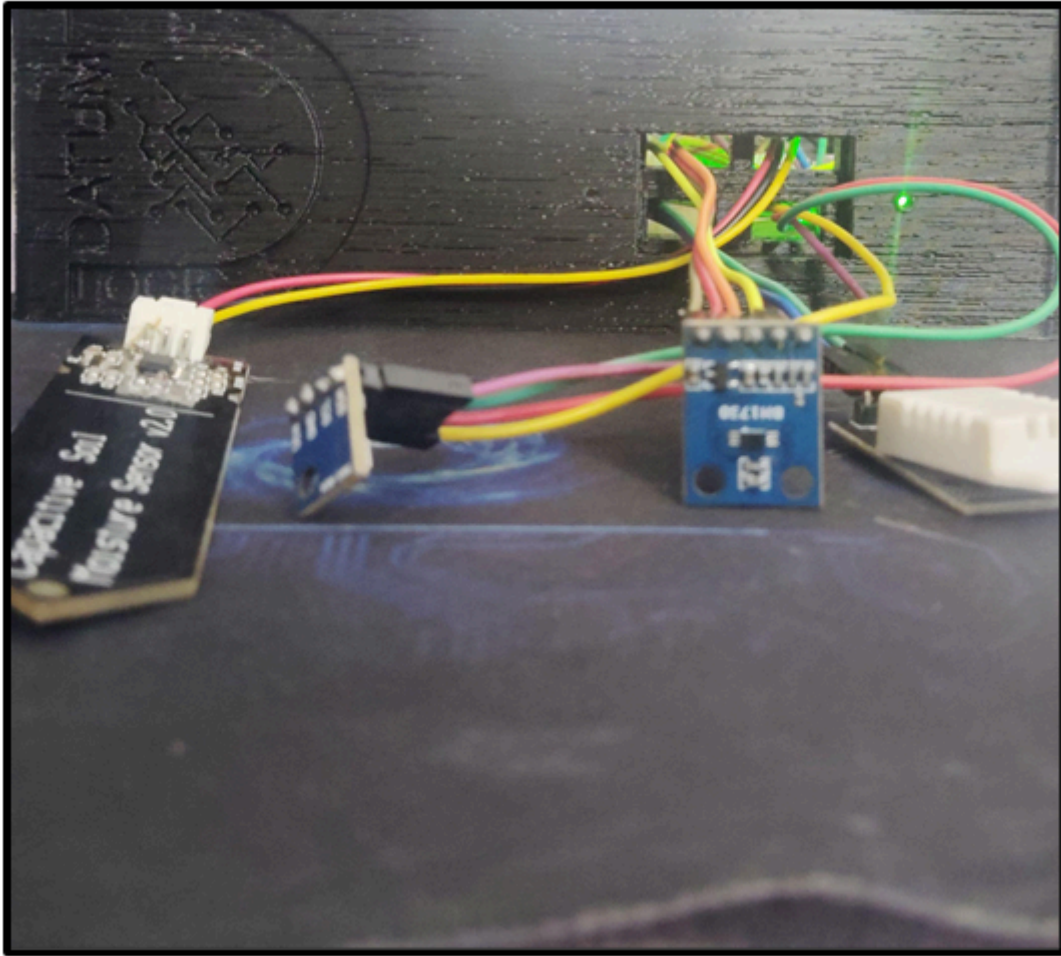
MAIN PCB



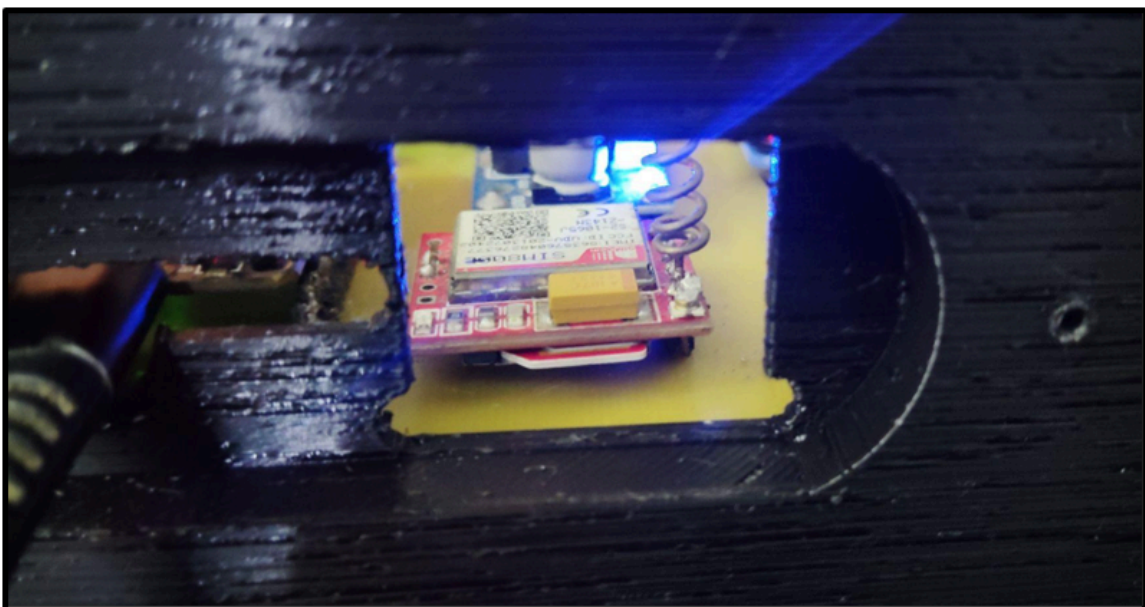
User Interface PCB



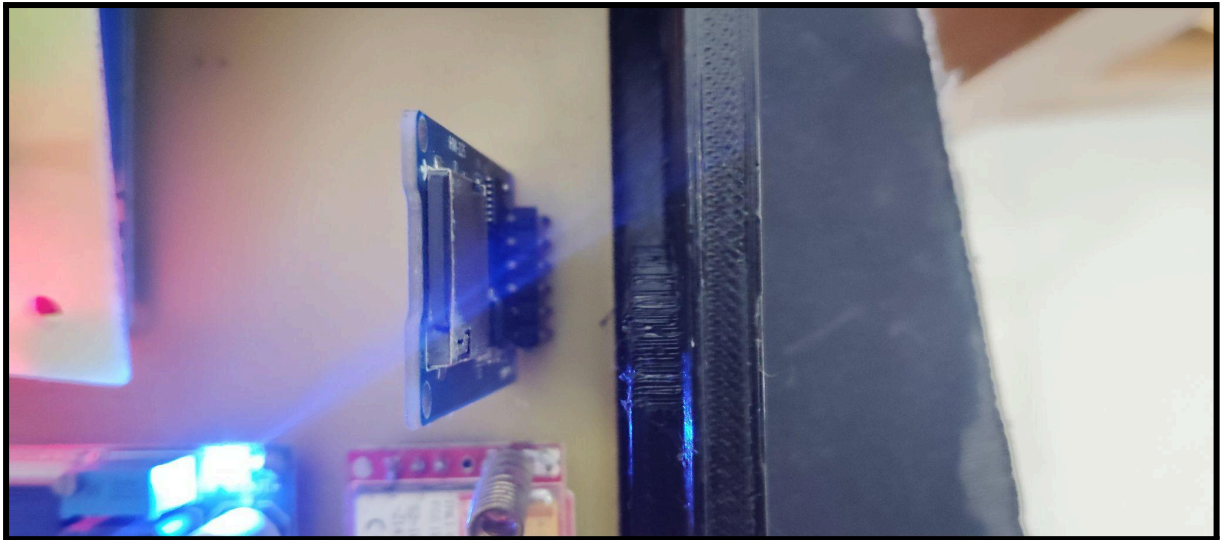
Power Jack and Switch



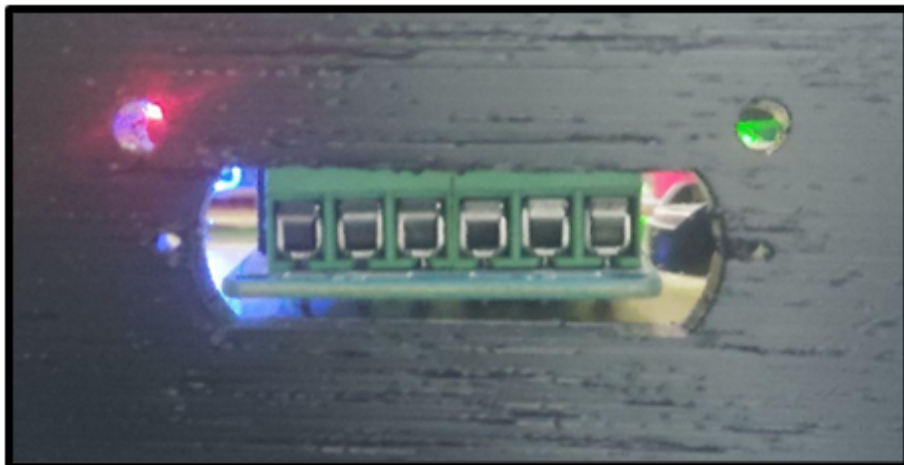
Sensors(From Left(SoilMoistureSensor, PressureSensor, LightIntensitySensor, Integrated Temperature and HumiditySensor))



GSM Module Access Slot



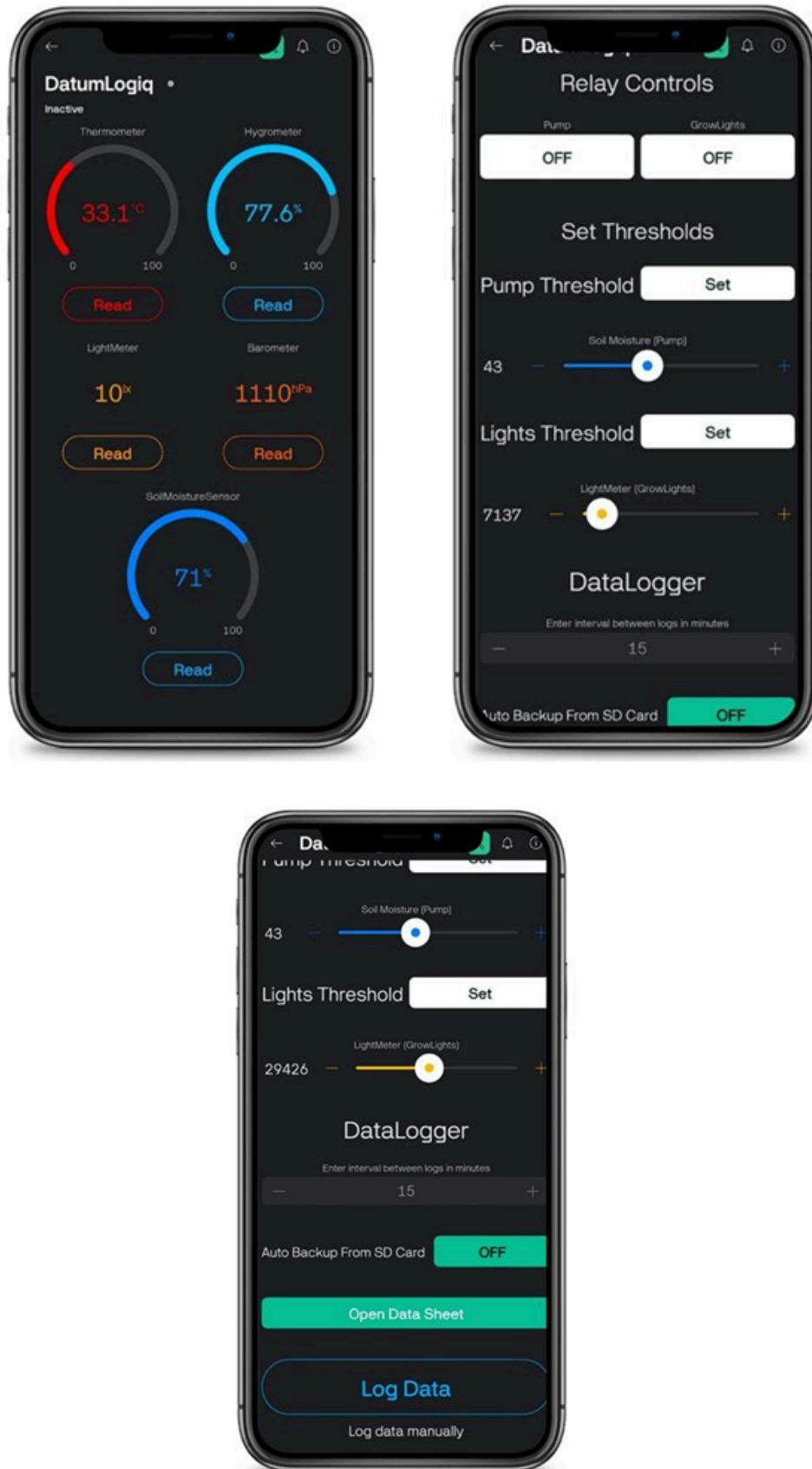
SD Card Module



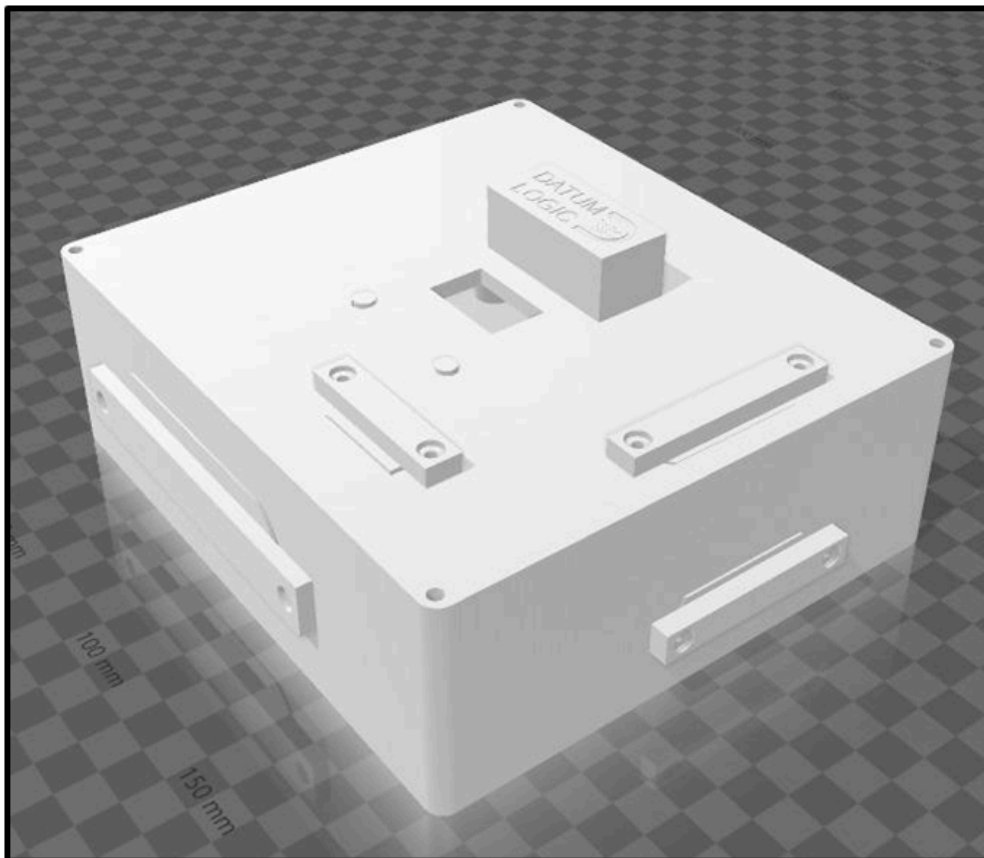
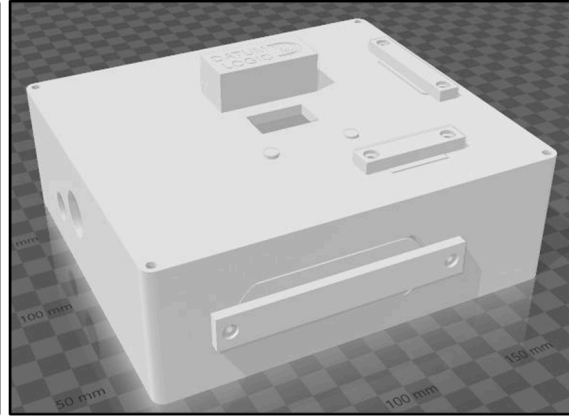
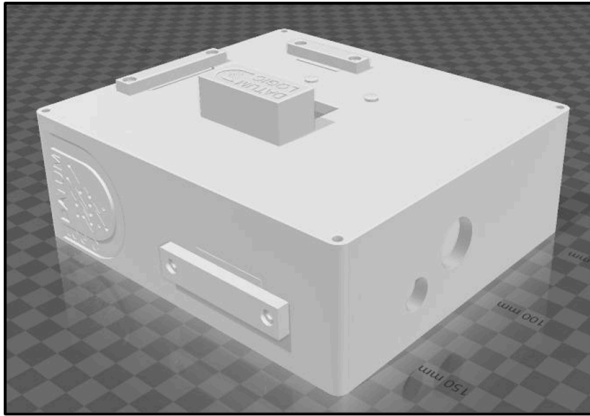
2 Channel Relay Module



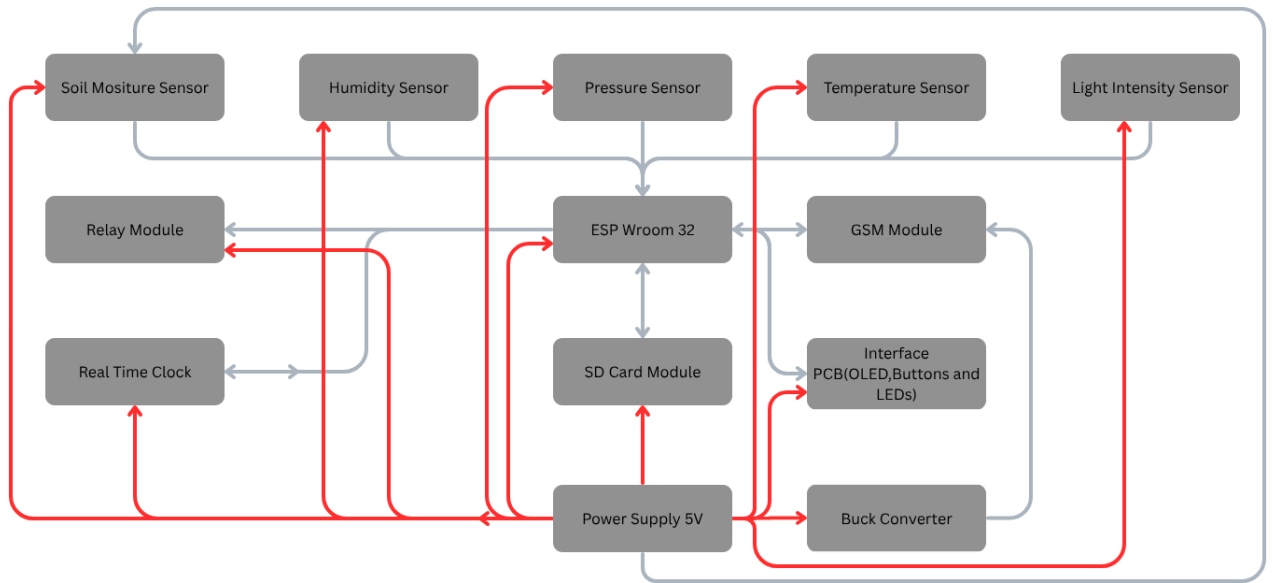
4x Connector Slots for External Sensors



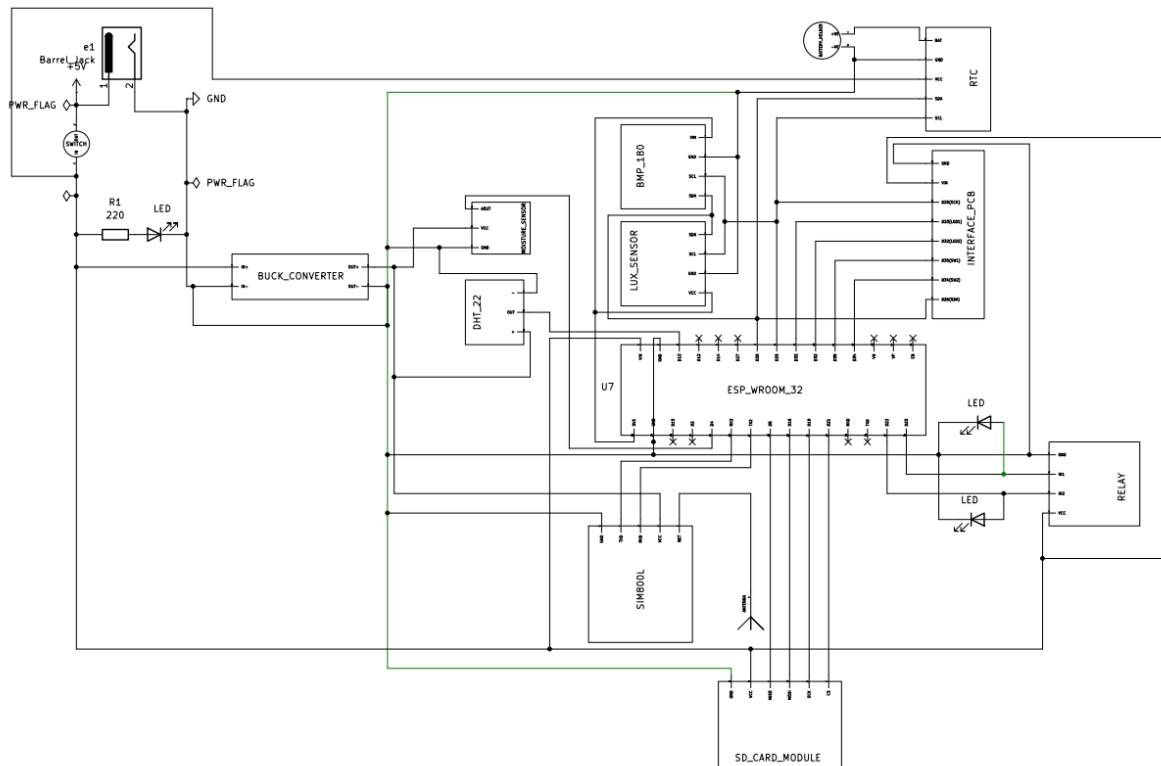
IoT Based Application for monitoring and Controlling the System



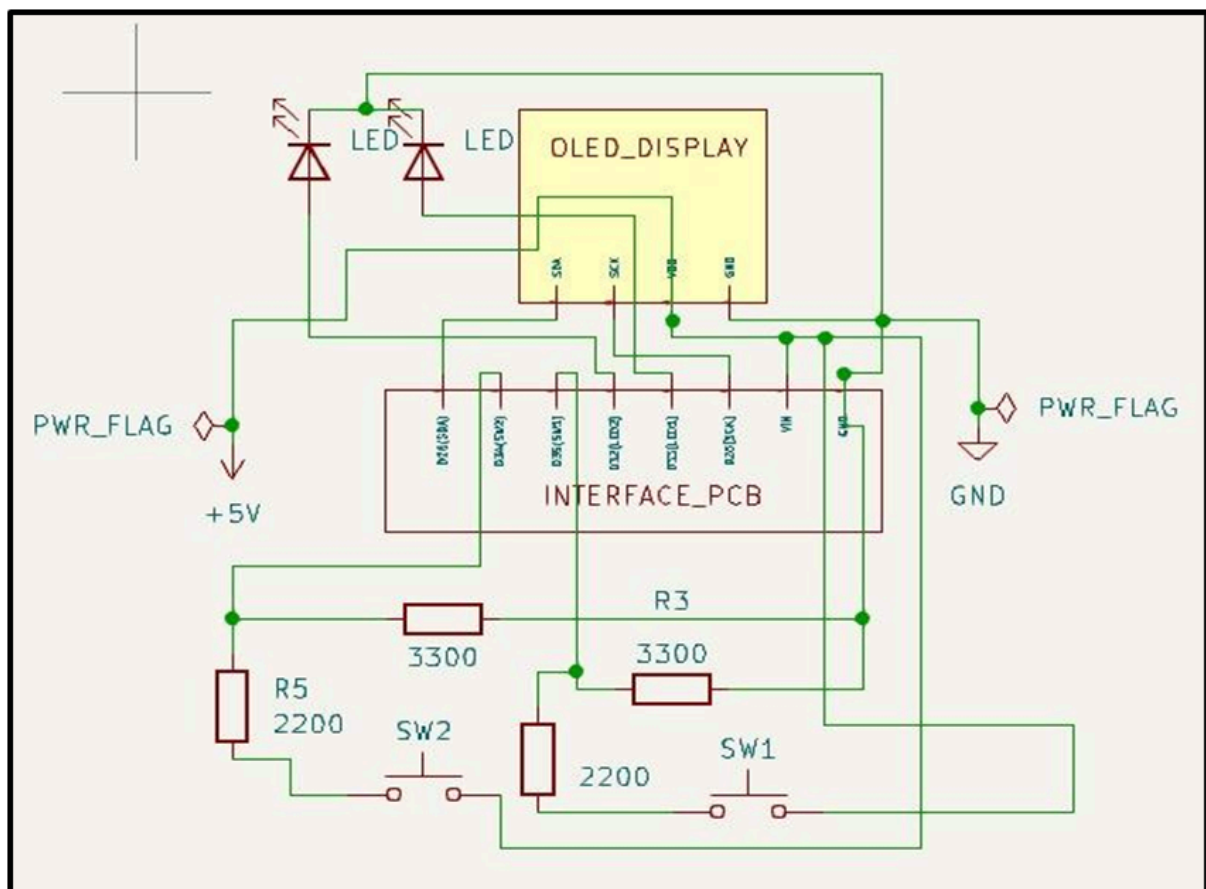
Different POVs of 3D model of the enclosure

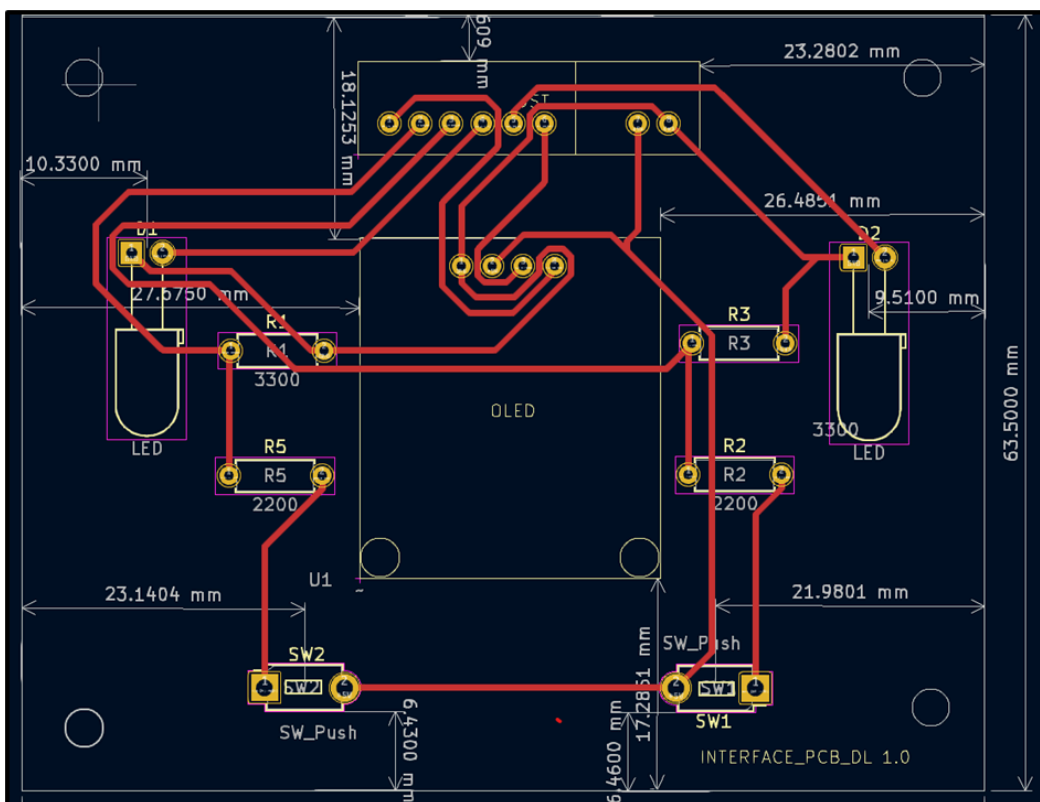
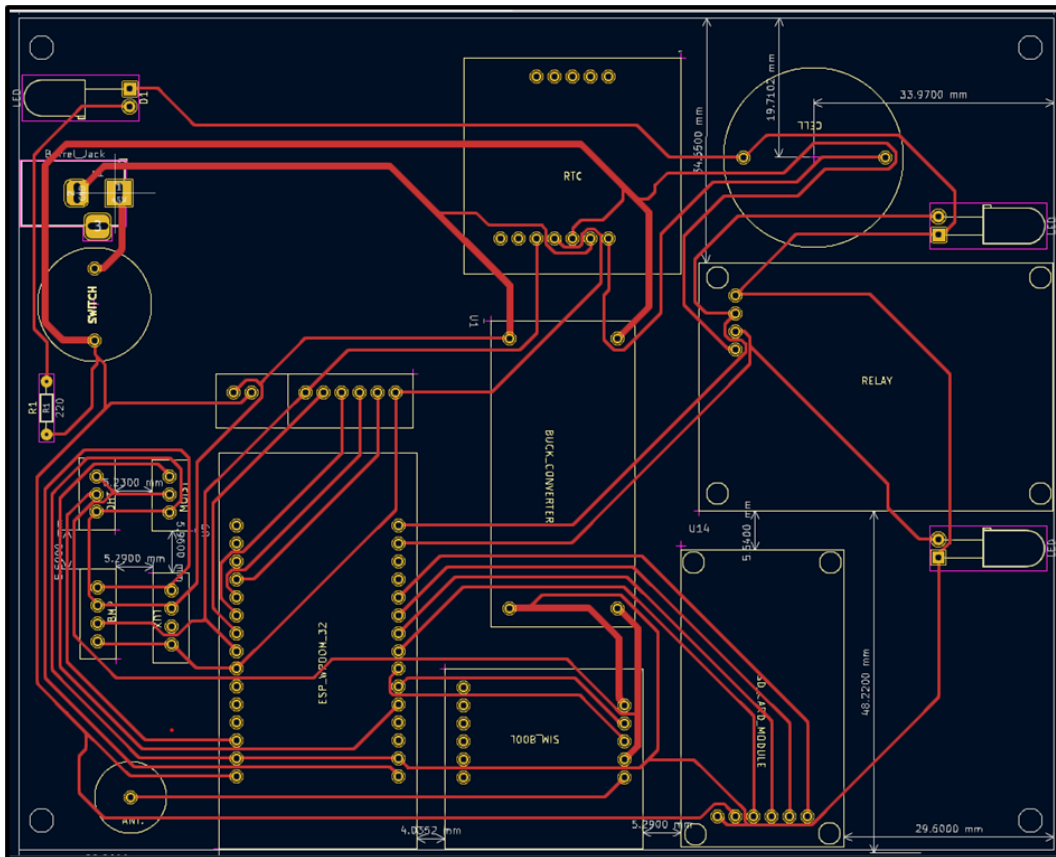


Block Diagram of the System



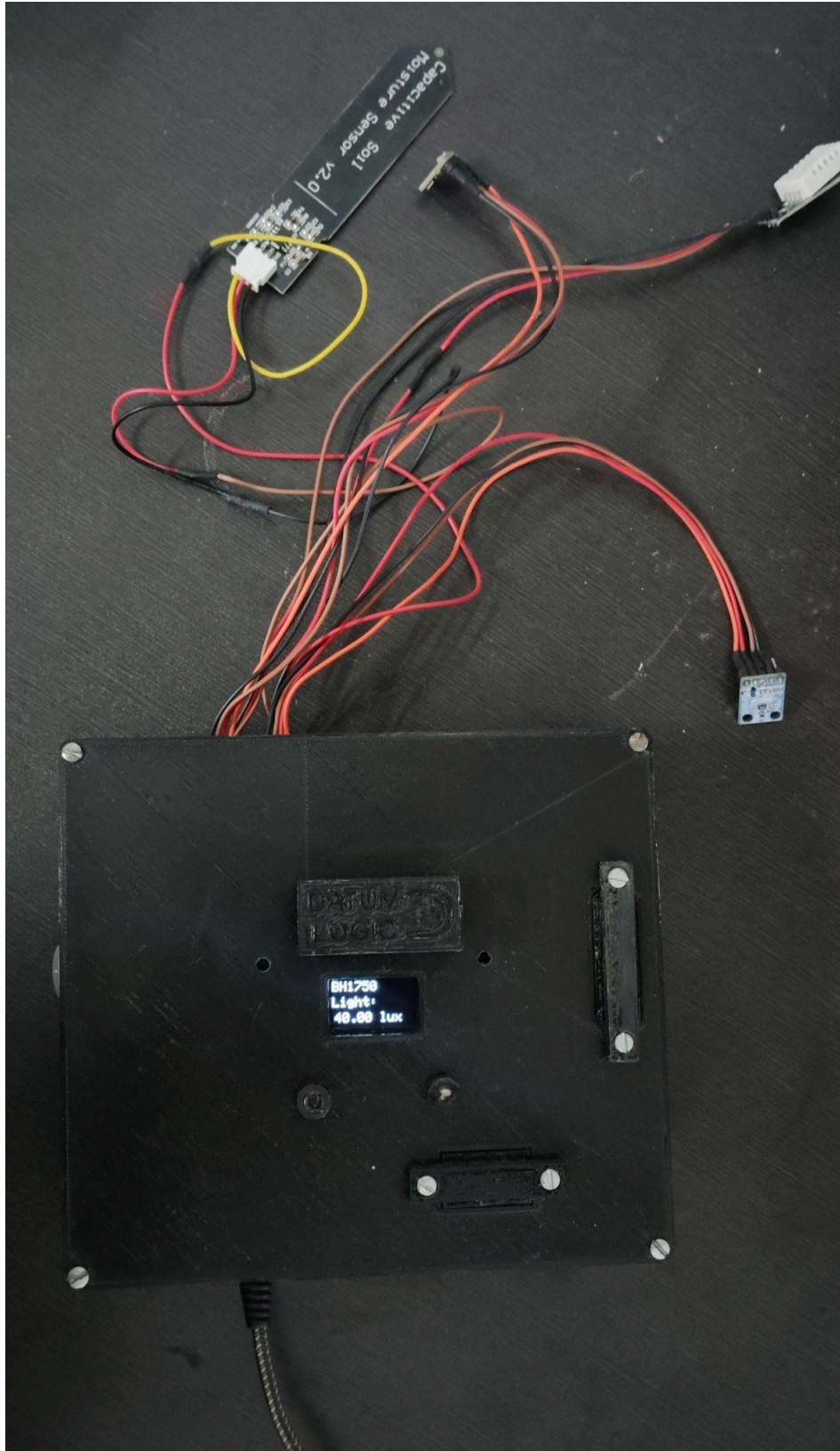
Main PCB Circuit Diagram





Design of Main PCB and Interface PCB

4.Final Product Image



5. Test Results

The following tests were conducted to verify the system's core functionalities:

1. Automation Logic: Pump and light triggers responded correctly to threshold values.
2. Network Fallback: Verified GSM takeover when WiFi was disabled.
3. Cloud Logging: Data correctly reflected on Google Sheets with expected intervals.
4. SD Backup: Local logs maintained even when internet connectivity was lost.
5. Blynk app: Real-time access to all sensor data is provided through the Blynk app.

These results confirm the system operates reliably under varying environmental and network conditions.

1	Date	Time	Temperature	Humidity	SoilMoisture	Atm.Pressure	LightIntensity
7033	31/12/24	23:06:42	31.5	76.1	71	1009.5	8.33
7034	31/12/24	23:13:22	31.5	75.3	71	1009.64	9.17
7035	31/12/24	23:50:48	31.5	76.1	71	1009.5	8.33
7036	31/12/24	23:50:49	31.5	75.3	71	1009.64	9.17
7037	31/12/24	23:50:49	31.3	72.3	72	1009.09	9.17
7038	01/01/25	0:05:47	31.2	72.2	71	1009.18	4.17
7039	01/01/25	0:20:47	31.2	72.6	71	1008.94	4.17
7040	01/01/25	0:35:47	31.2	73.6	71	1008.82	4.17
7041	01/01/25	0:50:48	31.1	74.5	71	1008.69	4.17
7042	01/01/25	1:05:49	31	75	71	1008.59	4.17
7043	01/01/25	1:20:47	31.1	74.3	71	1008.44	4.17
7044	01/01/25	1:35:48	31.1	74.3	71	1008.44	3.33
7045	01/01/25	1:50:48	31.1	74.3	71	1008.28	3.33
7046	01/01/25	2:05:49	31.1	75	71	1007.94	3.33
7047	01/01/25	2:20:48	31.1	74.9	71	1007.56	4.17
7048	01/01/25	4:03:29	30.7	73.2	71	1007.21	1.67
7049	01/01/25	4:12:55	30.6	73.2	71	1007.19	2.5
7050	01/01/25	4:12:55	30.6	73.2	71	1007.19	2.5
7051	01/01/25	4:21:01	30.6	73.4	71	1007.21	1.67
7052	01/01/25	4:21:10	30.6	73.3	71	1007.21	1.67
7053	01/01/25	4:21:18	30.6	73.3	71	1007.19	1.67
7054	01/01/25	4:21:23	30.6	73.3	72	1007.22	1.67
7055	01/01/25	15:38:39	33.8	67.4	71	1005.82	7.5
7056	01/01/25	15:55:05	28.8	66	72	1005.69	5.83
7057	01/01/25	16:07:51	27	61	71	1005.67	5.83
7058	01/01/25	16:17:51	26.4	59.9	72	1005.7	5
7059	01/01/25	16:27:48	26.1	60.5	72	1005.62	5.83
7060	01/01/25	16:37:51	30.6	62.7	71	1005.78	6.67
7061	01/01/25	16:47:50	31.1	64.8	71	1006.01	5.83
7062	01/01/25	16:57:51	31.3	65.6	71	1006.21	5
7063	01/01/25	17:07:56	31.4	66.6	71	1006.17	4.17
7064	01/01/25	17:17:53	31.4	67.4	71	1006.4	2.5
7065	01/01/25	17:27:53	31.5	67.3	71	1006.53	1.67
7066	01/01/25	17:37:54	31.6	66.3	71	1006.73	0.83

Sample of Data Logged to the Cloud

dataLog		+				...		
	A	B	C	D	E	F	G	H
173	Date	Time	Temperature	Humidity	Moisture	Pressure	LightIntensity	
174	31/12/2024	19:07:12	33.1	78.7	71	1008.39	7.5	
175	31/12/2024	19:21:17	32.9	78.7	71	1008.49	8.33	
176	31/12/2024	19:27:04	32.7	79.1	71	1008.6	7.5	
177	31/12/2024	19:27:20	32.7	78.9	71	1008.58	8.33	
178	31/12/2024	19:27:26	32.8	78.9	71	1008.65	8.33	
179	31/12/2024	19:28:22	32.8	78.6	71	1008.58	8.33	
180	31/12/2024	19:31:51	32.6	79	71	1008.81	8.33	
181	31/12/2024	19:32:02	32.7	79	71	1008.76	8.33	
182	31/12/2024	19:32:05	32.7	79	71	1008.68	8.33	
183	31/12/2024	19:32:07	32.7	79	71	1008.71	8.33	
184	31/12/2024	19:32:10	32.7	78.9	71	1008.68	8.33	
185	31/12/2024	19:32:11	32.7	79	71	1008.77	8.33	
186	31/12/2024	19:32:13	32.7	79	71	1008.75	8.33	
187	31/12/2024	19:32:14	32.7	79	71	1008.7	8.33	
188	31/12/2024	19:32:16	32.7	79	71	1008.76	8.33	
189	31/12/2024	19:32:18	32.7	79	71	1008.79	8.33	
190	31/12/2024	19:32:32	32.7	78.9	100	1008.7	8.33	
191	31/12/2024	19:32:34	32.7	78.8	100	1008.73	8.33	
192	31/12/2024	19:32:38	32.7	78.8	100	1008.67	8.33	
193	31/12/2024	19:32:46	32.7	78.9	71	1008.78	6.67	
194	31/12/2024	19:32:50	32.7	78.9	71	1008.78	8.33	
195	31/12/2024	19:32:51	32.7	78.9	71	1008.75	8.33	
196	31/12/2024	19:32:53	32.7	78.9	71	1008.78	8.33	
197	31/12/2024	19:32:54	32.7	78.9	71	1008.8	8.33	
198	31/12/2024	19:32:55	32.7	78.8	71	1008.83	8.33	
199	31/12/2024	19:32:57	32.7	78.8	71	1008.75	8.33	
200	31/12/2024	19:32:58	32.7	78.8	71	1008.69	8.33	
201	31/12/2024	19:32:59	32.7	78.9	71	1008.74	8.33	
202	31/12/2024	19:34:34	32.6	79.3	71	1008.82	7.5	
203	31/12/2024	19:36:19	32.7	79	71	1008.82	6.67	
204	31/12/2024	19:36:34	32.8	79	71	1008.77	6.67	
205	31/12/2024	19:48:59	32.9	79	71	1009.02	8.33	
206	31/12/2024	20:03:59	32.9	79.4	71	1009.08	8.33	
207	31/12/2024	20:04:56	32.9	79.5	71	1009.05	8.33	
208	31/12/2024	20:04:58	32.9	79.4	71	1009.01	8.33	
209	31/12/2024	20:04:58	32.9	79.4	71	1009.1	8.33	

Sample of Data Logged Locally

6. Conclusion

The ESP32-based environmental data logger is a high-featured, robust, and scalable solution for real-time monitoring and automated control. With dual-mode network support, strong local logging, and complete IoT integration, the system guarantees continuous operation and accessibility. Its modular design and remote configurability make it perfectly suited to various real-world applications.

7. References

- Espressif Systems. "ESP32 Technical Reference Manual." <https://docs.espressif.com>
- Blynk IoT Platform Documentation. <https://docs.blynk.io>
- Google Apps Script Guides. "Web Apps." <https://developers.google.com/apps-script/guides/web>